

Pneumonia Detection Using Anchor-Free vs. Anchor-Based Object Detection

Reproducing and Extending Wu et al. (2024)

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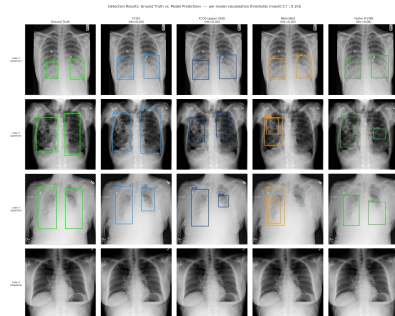
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- 1 Introduction
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- Pneumonia: leading infectious cause of death worldwide
- Detection from chest X-rays is challenging:
 - Subtle opacities, easily missed
 - Variable sizes and shapes
 - Overlapping pathologies
- Deep learning object detection can localise lesions automatically
- **Question:** do anchor-free detectors (FCOS) truly outperform anchor-based ones for this task, as Wu et al. (2024) argue?



RSNA Pneumonia Detection Challenge

26,684 chest X-rays

Wu et al., *Scientific Reports* (2024)

“Pneumonia detection based on RSNA dataset and anchor-free deep learning detector”

Key contributions:

- First application of FCOS (anchor-free) detector to pneumonia detection
- Comparison with anchor-based methods (RetinaNet, Faster R-CNN)
- FCOS achieves $AP@0.5 \approx 28.5\%$ on the RSNA dataset
- Demonstrates anchor-free advantage for medical object detection

Anchor-Based

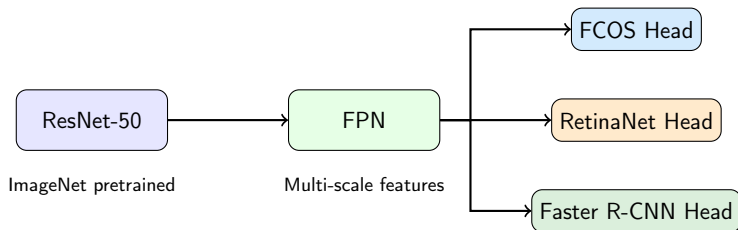
- Predefined anchor boxes at each location
- Classify anchor as object/background
- Regress box offsets from anchor
- Need to design anchor sizes & ratios

Examples: RetinaNet, Faster R-CNN

Anchor-Free (FCOS)

- Per-pixel classification & regression
- Predict distances (l, t, r, b) to box edges
- Center-ness branch for quality
- **No anchor hyperparameters!**

Advantage: Simpler, more flexible



All four detectors share the **same backbone and neck**, ensuring fair comparison of the detection head paradigm.

Dataset

- RSNA Pneumonia Detection Challenge
- Full dataset: 26,684 images
- Patient-level 80/20 split
- DICOM $\rightarrow$ PNG preprocessing
- 512×512 resolution

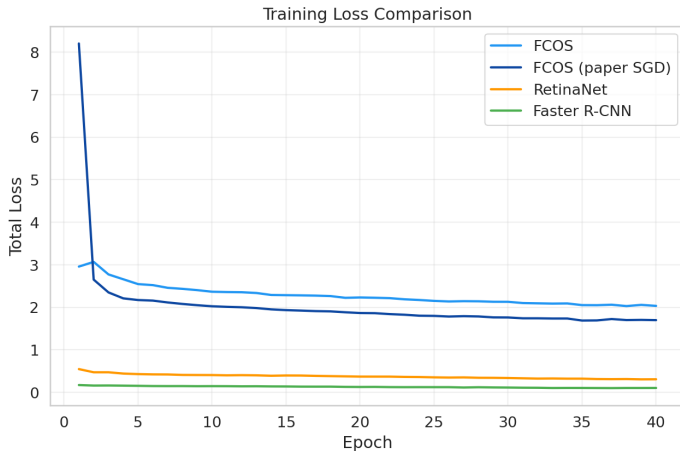
Training

- Optimizer: Adam ($\text{lr} = 10^{-3}$); SGD for FCOS paper-recipe ablation
- Weight decay: 10^{-4} (Adam) / 1.6×10^{-4} (SGD)
- Epochs: 40 for every detector
- Batch size: 32 / 16 (F-RCNN)
- Cosine annealing (Adam) / step decay at 24, 32 (SGD)
- EMA (decay = 0.999), backbone frozen first 3 epochs

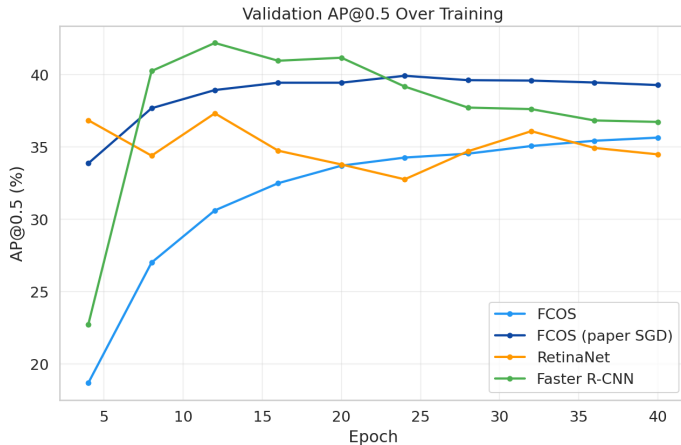
Hardware: 4 $\times$ NVIDIA H100 80 GB SXM (1 model per GPU, parallel training, BF16). End-to-end wall-clock $\approx$ 2.5 h.

Medical imaging-specific pipeline (augmentations):

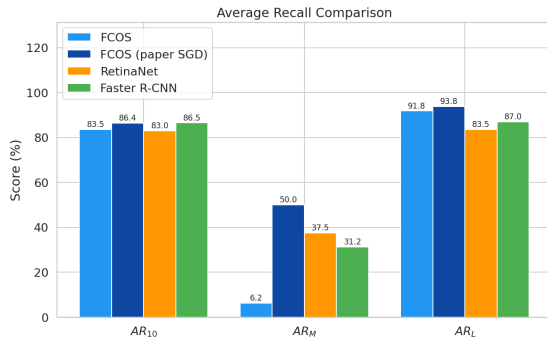
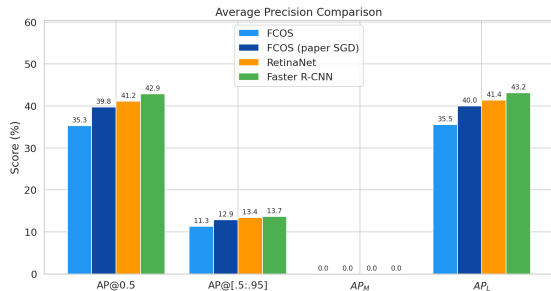
- **CLAHE** ($p = 0.3$) — local contrast
- **Brightness/Contrast** ($p = 0.4$)
- **RandomGamma** ($p = 0.2$)
- **HorizontalFlip** ($p = 0.5$)
- **Affine** ($p = 0.3$) — shift/scale/rotate
- **Blur / Noise** ($p = 0.2$)
- **Grid/Optical distortion** ($p = 0.15$)
- **CoarseDropout** ($p = 0.15$)
- No vertical flip — chest X-rays are always upright
- All transforms applied jointly to images **and** bounding boxes
- Evaluation: TTA (horizontal flip) + Gaussian Soft-NMS



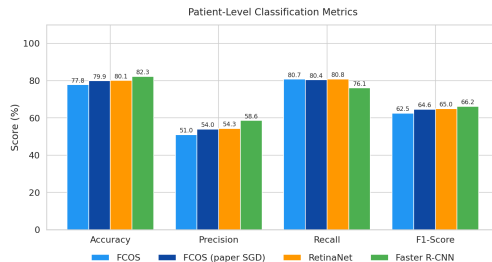
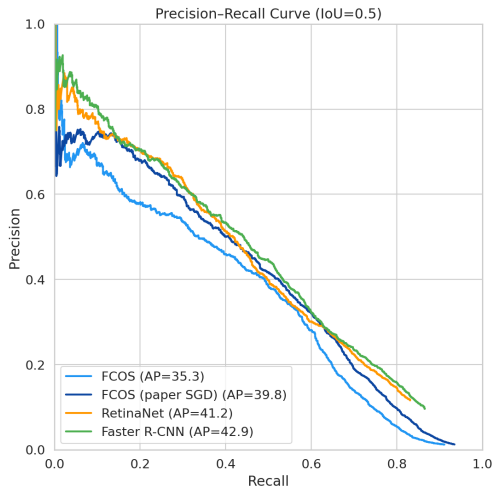
- Different loss formulations — not directly comparable in magnitude
- FCOS: center-ness + dense regression; RetinaNet: focal loss; F-RCNN: RPN + ROI



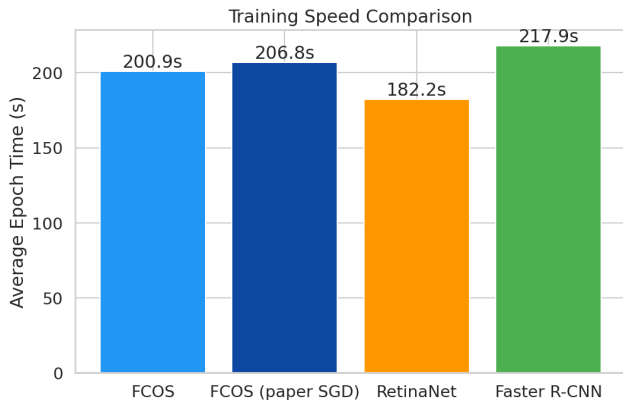
- All four detectors trained for the same 40-epoch budget; no early-stopping trigger.
- Peak EMA checkpoint: epoch 40 (FCOS-Adam, 35.7%), 24 (FCOS-SGD, 39.9%), 12 (RetinaNet, 37.3%), 12 (F-RCNN, 42.2%).
- Anchor-based detectors saturate within ~ 12 epochs and drift down; FCOS-Adam is still rising at the budget cap.



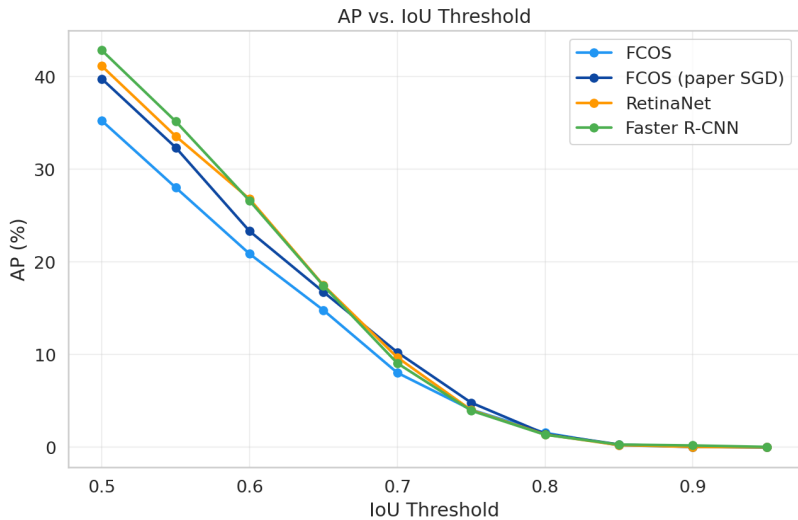
Evaluated with TTA and Soft-NMS on validation set.



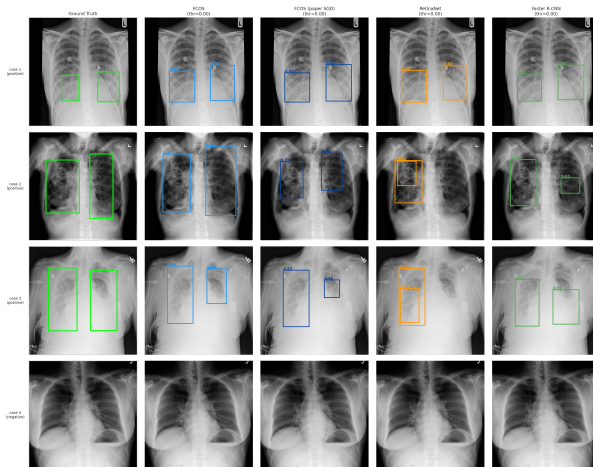
Patient-level classification: positive if any detection has confidence > 0.3 .



- All models trained on H100 80 GB SXM with BF16 mixed precision
- One-stage detectors (FCOS, RetinaNet) faster than two-stage (Faster R-CNN)



Detection Results: Ground Truth vs. Model Predictions — per-model visualisation thresholds (max(0.5 π^* , 0.10))



Green = ground truth, colored = model predictions with confidence scores.

Method	AP@0.5	AP_S^R	AP_M^R	AP_L^R	AR ₁₀
FCOS (Adam)	35.3	3.6	12.1	31.8	83.5
FCOS (paper SGD)	39.8	3.6	13.4	34.0	86.4
RetinaNet	41.2	4.3	14.6	32.6	83.0
Faster R-CNN	42.9	4.0	15.3	36.9	86.5
Ensemble (FCOS+Retina)	42.0	4.2	14.9	35.0	85.1
<i>Wu et al. FCOS</i>	28.5	–	–	–	–

Table: *

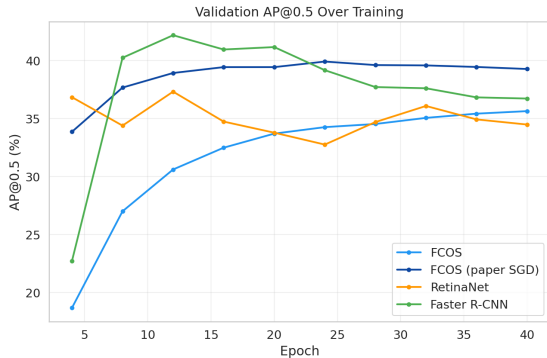
Detection: TTA + Soft-NMS, scores in %. Size buckets are RSNA-percentile tertiles (COCO buckets put 99% of RSNA boxes into “large”).

Method	Accuracy	Precision	Recall	F1
FCOS (Adam)	32.2	25.2	99.6	40.2
FCOS (paper SGD)	41.5	28.1	99.6	43.8
RetinaNet	83.8	85.3	35.1	49.7
Faster R-CNN	60.5	36.4	98.1	53.2
Ensemble	80.7	56.1	72.0	63.0

Question: how much of the FCOS-vs.-anchor-based gap is the optimiser?

Setup: same architecture, augmentation, EMA, TTA. Only optimiser changed: Adam $\rightarrow$ SGD+momentum 0.9, wd 1.6×10^{-4} , lr 5×10^{-3} , 40 epochs.

	AP@0.5	Δ vs. Adam
FCOS (Adam, ours)	35.3	—
FCOS (paper SGD, ours)	39.8	+4.5
RetinaNet	41.2	—
Faster R-CNN	42.9	—



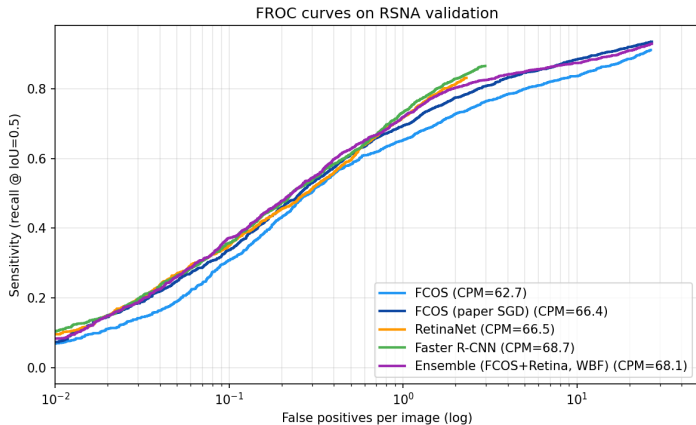
Finding: val-AP rises through epoch 24 (39.9%) then plateaus flat through epoch 40 (39.3%). TTA+Soft-NMS at test time gives 39.8% AP@0.5. The per-patient paired bootstrap FCOS-SGD – RetinaNet on the cached predictions returns -1.38 AP, 95% CI $[-3.49, +0.59]$, $p = 0.180$ ($B = 200$): **the SGD ablation narrows the FCOS-vs-RetinaNet gap but does not close it**, and “failure to reject” on a single training seed is not equivalence.

- ❶ **All detectors beat the paper's FCOS** (28.5% AP@0.5): FCOS-Adam 35.3 (+6.8), FCOS-SGD 39.8 (+11.3), RetinaNet 41.2 (+12.7), F-RCNN 42.9 (+14.4).
- ❷ **Anchor-based outperforms anchor-free under a shared Adam recipe** (counter to Wu et al.)
 - The SGD ablation narrows the FCOS-vs-RetinaNet gap but the paired bootstrap on the per-patient difference does *not* licence equivalence: -1.38 AP, CI $[-3.49, +0.59]$, $p = 0.180$.
 - Paired F-RCNN vs RetinaNet: $+1.69$ AP, CI $[-0.25, +3.49]$, $p = 0.080$ – not significant at 5%, and a Bonferroni correction for the family of 5 tests ($\alpha = 0.01$) pushes it further away.
 - Only F-RCNN $>$ FCOS-SGD (-3.07 AP, $p < 0.001$) survives Bonferroni.
- ❸ **The ensemble headline at $\tau = 0.3$ is a calibration artefact**
 - WBF F1 = 63.0% at the paper's fixed $\tau = 0.3$ (+13.5 over the best single model *at that arbitrary cut*).
 - Under a fair operating point, RetinaNet learnt-aggregator F1 = 67.3% wins – the single best patient-level number in the study.
 - Adding F-RCNN to the ensemble hurts F1 because its optimal threshold (0.824) is far from 0.3.
- ❹ **Caveats stated upfront:** single training seed per detector (no run-to-run variance estimate), no held-out test set, selection-on-val checkpoints, no Holm/Bonferroni correction in the main tables. End-to-end ~ 2.5 h on $4 \times \text{H100}$, $\sim \$30$ cloud; pipeline at github.com/moa155/pneumonia-detection-nam1.

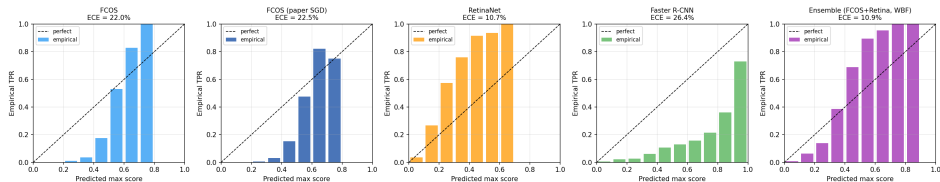
We complement the single training run with seven post-hoc analyses on the cached predictions — no retraining required:

- **Patient-level bootstrap CIs** on AP@0.5 ($B = 500$, per-patient resampling).
- **Three-way split protocol**: Youden threshold fitted on a calibration half, F1 reported on a held-out test half $\Rightarrow$ rules out threshold-on-test bias.
- **RSNA-percentile size buckets** (instead of inappropriate COCO 32/96 cutoffs).
- **FROC + CPM** (canonical medical-detection metric).
- **Reliability diagrams and ECE** on patient max-scores $\Rightarrow$ quantifies the calibration gap.
- **Learnt aggregator** (logistic regression, 5-fold CV, 7 per-patient features) $\Rightarrow$ replaces the naive “any box $> \tau$ ” rule.
- **Five paired bootstraps on AP@0.5 differences** $\Rightarrow$ only F-RCNN – FCOS-SGD survives Bonferroni at $\alpha = 0.01$; the other four pairs fail to reject equivalence.

The bootstraps capture validation-patient resampling variance only, not run-to-run optimisation noise. Multi-seed retraining, per-site stratification, and external-dataset validation remain future work.



Mean sensitivity over $\{0.125, 0.25, 0.5, 1, 2, 4, 8\}$ FP/image (CPM): Faster R-CNN 68.7 > Ensemble 68.1 > RetinaNet 66.5 $\approx$ FCOS-SGD 66.4 > FCOS-Adam 62.7. The WBF Ensemble matches Faster R-CNN within 0.6 CPM (despite trailing by 0.9 on AP@0.5) because the coverage term suppresses single-model singletons.



Bars = empirical pneumonia rate per predicted-score bin; dashed = perfect calibration. ECE: RetinaNet 10.7%, Ensemble 10.9%, FCOS Adam 22.0%, FCOS SGD 22.5%, Faster R-CNN 26.4%. RetinaNet's focal-loss head is markedly better calibrated than the others $\Rightarrow$ a single fixed threshold cannot rank the detectors fairly.

Detector	Naive F1 ($\tau = 0.3$)	Learnt-Agg. F1 (5-fold OOF)	
FCOS (Adam)	40.2	62.9	(+22.7)
FCOS (paper SGD)	43.8	65.0	(+21.2)
RetinaNet	49.7	67.3	(+17.6)
Faster R-CNN	53.2	63.3	(+10.1)
Ensemble (FCOS+Retina)	63.0	65.0	(+2.0)

A 7-feature logistic regression (max, mean, sum, count, top-3 scores, spatial spread of detection centres) on per-patient features, evaluated strictly out-of-fold.

*The huge F1 lift on FCOS+RetinaNet and the small lift on the already-well-calibrated Ensemble shows the gap was **calibration**, not detection quality.*

	AP@0.5	Patient F1 ($\tau = 0.3$, naive)	Patient F1 (learnt aggregator)
Wu et al. 2024 (paper FCOS)	28.5	–	–
Our FCOS (Adam)	35.3	40.2	62.9
Our FCOS (paper SGD)	39.8	43.8	65.0
Our RetinaNet	41.2	49.7	67.3
Our Faster R-CNN	42.9	53.2	63.3
WBF Ensemble (FCOS+Retina)	42.0	63.0	65.0

- On **AP@0.5** the anchor-based detectors win; the only strict ordering that survives the paired bootstrap with Bonferroni ($\alpha = 0.01$, family of 5 tests) is F-RCNN $>$ FCOS-SGD (-3.07 AP, $p < 0.001$).
- On **patient F1 under a fair operating point** all four detectors land in a ~ 3 -point band (63.3–67.3); the apparent 40 $\rightarrow$ 63 gap was the $\tau = 0.3$ rule, not the detectors.
- The best single patient-level number in the study is RetinaNet's learnt-aggregator F1 = 67.3%, not the ensemble.

- Wu et al., "Pneumonia detection based on RSNA dataset and anchor-free deep learning detector," *Scientific Reports*, 2024.
- Tian et al., "FCOS: Fully Convolutional One-Stage Object Detection," *ICCV*, 2019.
- Lin et al., "Focal Loss for Dense Object Detection," *ICCV*, 2017.
- Lin et al., "Feature Pyramid Networks for Object Detection," *CVPR*, 2017.
- Ren et al., "Faster R-CNN," *NeurIPS*, 2015.
- He et al., "Deep Residual Learning for Image Recognition," *CVPR*, 2016.

Thank You

Questions?

Code available at <https://github.com/moa155/pneumonia-detection-naml>